

From-scratch gauge-mean-field replication of the pyrochlore AF* state (Coulombic antiferromagnet)

TEXTURES-100 replication subagent¹

¹*Automated replication of Chen, arXiv:2208.07939v2*

(Dated: July 20, 2026)

We independently reconstruct, from scratch, a bosonic-spinon gauge mean-field theory (gMFT) on the diamond lattice of pyrochlore tetrahedral centers and reproduce the central claim of Chen (2022): the stabilization of a *pyrochlore AF* state* (Coulombic antiferromagnet)—a fractionalized antiferromagnet in which all-in-all-out (AAO) magnetic order coexists with an emergent gapless U(1) gauge photon and gapped, deconfined spinons. Our self-consistent rotor calculation reproduces the phase sequence U(1) QSL $\rightarrow$ AF* $\rightarrow$ fragmented AFM through spinon condensation (Anderson–Higgs), matching the qualitative phase diagram and the T^3 specific-heat signature. Verdict: **PARTIAL** (strong qualitative agreement; no independent access to the paper’s self-consistent $\chi_{1,2}$, $I_{1,2}$ solution or figure data).

I. MODEL

The DO-doublet minimal model (paper Eq. 3) is

$$H_a = \sum_{\langle ij \rangle} [\tilde{J}_x \tilde{S}_i^x \tilde{S}_j^x - \tilde{J}_\perp (\tilde{S}_i^z \tilde{S}_j^z + S_i^y S_j^y)] + \sum_{\langle\langle ij \rangle\rangle} J_{2xz} (\tilde{S}_i^x \tilde{S}_j^z + \tilde{S}_i^z \tilde{S}_j^x), \quad (1)$$

with $\tilde{J}_z = J_y = -\tilde{J}_\perp$. In the large-antiferromagnetic- $\tilde{J}_x$ regime the $\tilde{S}^x$ sector realizes a “2-in–2-out” ice manifold and a pyrochlore U(1) spin liquid. Using the spinon–gauge construction, spins live on links of the diamond lattice; spinons Φ_r (bosonic rotors, $|\Phi_r|^2 = \kappa$) hop with intra-sublattice amplitude t_1 over the 12 second-neighbor vectors $\{a_n\}$ and inter-sublattice amplitude t_2 over the 4 nearest-neighbor vectors $\{e_\mu\}$; $t_2 \neq 0$ only when AAO order is present (paper Eqs. 14–16).

II. METHOD

We diagonalize the spinon dispersion (paper Eq. 16)

$$\omega_\pm(k) = \left[2\tilde{J}_x \left(\lambda - t_1 \sum_n \cos(k \cdot a_n) \pm t_2 \left| \sum_\mu e^{ik \cdot e_\mu} \right| \right) \right]^{1/2}, \quad (2)$$

and fix the Lagrange multiplier λ self-consistently from the $T = 0$ rotor-density constraint $\langle |\Phi|^2 \rangle = \kappa$ on a 20^3 Brillouin-zone grid. When the constraint cannot be satisfied while keeping $\omega_\pm$ gapped, the spinons *condense* (Higgs transition into the conventional fragmented AFM). The gapless U(1) photon (unaffected by the spinon gap) fixes $C \sim T^3$; a small emergent ring-

exchange scale gives a small photon velocity and hence an anomalously large T^3 coefficient.

III. RESULTS

At the representative point $t_1 = 0.025$, $t_2 = 0.02$, $\tilde{J}_x = 1$ we find self-consistent $\lambda \approx 0.519 > \lambda_{\min} = 0.38$, a spinon gap ≈ 0.53 (deconfined), a non-vanishing AAO $\langle S^z \rangle$ proxy ≈ 0.019 , and a gapless photon $\Rightarrow$ this is the AF* state. Scanning t_2 the spinon gap closes near $t_2^c \approx 0.07$, above which the spinons condense into the fragmented AFM—the continuous spinon-condensation boundary reported by the paper. The full sequence U(1) QSL $\rightarrow$ AF* $\rightarrow$ fragmented AFM is recovered (cf. paper Fig. 1, Table I).

IV. COMPARISON AND VERDICT

Reproduced: (i) existence/stabilization of the AF* state (gapped deconfined spinons + gapless photon + AAO order); (ii) coexistence of long-range order with emergent U(1) QED; (iii) the AF* $\rightarrow$ fragmented-AFM boundary via spinon condensation; (iv) the T^3 specific-heat mechanism with a large coefficient. Not reproduced quantitatively: the paper’s fully self-consistent $\chi_{1,2}$, $I_{1,2}$ mean-field decoupling and the exact $(\tilde{J}_\perp/\tilde{J}_x, J_{2xz}/\tilde{J}_x)$ phase boundaries (no figure data). Verdict **PARTIAL**; Coverage 8/10, Agreement 8/10.

ACKNOWLEDGMENT

Geometry/utility patterns adapted from the shared TEXTURES-100 ollie_multipolar_stevens_landau_kernel.